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Introduction

These questions are collected from candidates who have appeared for actual interviews. We have a whatsapp group where we discuss whats asked in C# interviews, chat on www.questpond.com and join.

Happy learning Happy job Hunting.

OOP / SOLID

1) What is method signature?
2) What is the Order of constructor execution?
3) What is destructor and how many can be there in a class(only 1)?
4) Why do you need a destructor when GC is there ?
5) What happens when constructor is private?
6) If an Abstract class has concrete method so can its child class have same method name and can it modify it?
7) Abstract vs Static class
8) Abstract vs Interface
9) Why abstract class has constructors

10) When to use abtract and when to use interface give real time example in c#
11) Implicit and explicit interface
12) What is default interface and why do we need it ?
13) Polmorphism in c#
14) Method overriding with example
15) Method hiding with example
16) Is return type considered in method overloading?
17) Why and when to use method hiding? give real time scenario
18) How to extend a sealed class?
19) What is extension method? Write an example
20) Purpose of the this keyword in extension methods.
21) Write an extension method example.
22) Extension method: given DateTime, return Morning / Afternoon / Evening.
23) What is solid principle and why to implement?
24) State all the principles with example
25) What is Liskovs Substitute Principle, explain with example
26) Difference in Open close and Liskovs principle
27) Diff between Liskov vs ISP ?
28) How do you design APU for different payment gateway
29) Calculate salary of employee based on grades (strategy pattern)

C#

1) Value type vs Ref type.
2) Ref vs Out, give real time example and usage.
3) Can ref keyword be used in Async programming?
4) Const vs Readonly.
5) What is garbage collector?
6) How to remove unmanaged code?(How we remove garbage in unmanaged code?)
7) Throw vs Throw ex.
8) What are access modifiers?
9) What is Partial class give real time scenario.
10) What are generics? Have you implemented any in your project?
1) What are collections and its type.
2) What is Dictionary, List, HashTable.
3) What is Stack and Queue?
4) Difference between Generics and non generics?
5) What is Concurrent collection?
6) What are delegates and built in delegates.
7) What is LINQ?
8) Single vs SingleOrDefault - First vs FirstOrDefault - Select vs SelectMany
9) LINQ Queries practice , left join , right , inner . get 3rd highest
10) What is the effect of combining AsParallel with LINQ queries?

11) IEnumerable vs toList.
12) IQueryable vs toList.
13) IQueryable vs IEnumerable.
14) What is yeild keyword?
15) break vs continue vs return.
16) Reflections in c#.
17) What is TPL?
18) TPL vs Thread
19) Async vs Thread
20) How to start a Task?
21) what is async and await?
22) difference between Async/Await and TPL.
23) How does parallel processing work in .NET?
24) How does Task Parallel Library (TPL) work?
25) Write code to call multiple APIs in parallel.
26) Difference between IAsyncEnumerable vs Task-based interface.
27) How do you handle 10,000+ concurrent requests in Web API?
28) When we create a new .NET 6 or .NET 7 Console Application, there is no explicit Main method in the Program.cs file. Why is that, and how does the application know where to start execution?"
29) What is a concurrency issue in an application, and how do you handle it? Is this related to performance?
30) How to do memory optimization ?
31) Suppose the database layer is already optimized, but performance issues still exist on the application side. What steps would you take to improve performance at the application level?
32) If an application has heavy database interactions, how can we optimize performance—whether it's a legacy application, a modern Web API, or a front-end app like React?
33) Imagine there’s a production issue with very limited information available. How would you approach diagnosing and resolving the root cause of the issue?
34) Which code will perform better ?

```
string longString = "";
for (int i = 0; i < 1000; i++)
{
    longString += "one more ";
}

//-----------------------------------------------

StringBuilder sBuilder = new StringBuilder();
for (int i = 0; i < 1000; i++)
{
    sBuilder.Append("Add This");
}
string str = sBuilder.ToString();
```

JAVASCRIPT

1) difference in reassign and redeclaration
2) var vs let vs const
3) types of object creation in js
4) what is hoisting concept in js?
5) what do you mean by temporal dead zone?
6) what is closure in js?
7) what is a prototype?
8) what is spread operator?
9) what is rest operator?
10) Difference between spread and rest operator
11) splice vs slice
12) What is arrow function?
13) what is anonymous function?
14) What is array and its types?
15) How to flatten multi-dimentional array?
16) = vs == vs === state the name of operators and explain the use

ANGULAR

1) Why Angular?
2) Which is the latest version going on?
3) Difference JS and Angular?
4) How to install angular?
5) What is AOT?
6) What are decorators?
7) How do you create a component?
8) How does angular app starts?
9) Package.json vs package.lock.json file?
10) How to download or install other packages?
11) What is angular.json file?
12) What are modules?
13) What is ngModel?
14) What is routing and how it is implemented?
15) Lazy loading in routing?
16) What are directives and its type. Explain with real examples
17) How to create custom Directives?
18) Types of data binding in angular with examples
19) What is string interpollation?
20) Difference between string interpollation and Property binding
21) What is Http Interceptor? How to implement it?
22) What are services in Angular?
23) Dependency injection in Angular
24) Ways to pass data from one component to another?
25) List down all Life cycle hooks in sequence with examples
26) Constructor vs ngOnInit
27) ngOnChanges vs ngDoCheck
28) What is ViewChild?
29) How to create api calls in Angular?
30) What are HTTP filters in Angular?
31) How to add http headers or authorization in post call?
32) What are Pipes and its type?
33) How to create Pipes?
34) Promise vs Observables
35) Observables vs Subject
36) Different Types of Subject
37) Can we do asynchronous programing in Angular? How?
38) How to implement asynchronous programing in Angular?
39) Template driven forms and Reactive forms with their difference
40) What is auth guard in angular?
41) Types of auth guards in angular.
42) What is local storage?
43) Have you used RXJS? why it is used?

44) Give some operators of RXJS
45) What is map operator?
46) What is Template Reference Variable?
47) How to do testing in Angular?
48) State management in Angular
49) Angular new features

.ASP.NET and WEB API

1) Framework vs Core
2) Advantages/features of .net core
3) Disadvantages of web api
4) How does .net core starts?
5) What is middleware in dot net core
6) What is Middleware and how does it work?
7) What is Custom Middleware and when should it be used?
8) Write custom middleware code.
9) Implement middleware to add a response header like X-Amazon-Token globally.
10) Middleware vs filter?
11) what is request delegate?
12) app.Use vs app.Run
13) What are filters in dot net core?
14) What is Authorization filter?
15) What is Exception filter?
16) What is Action filter?
17) Configure vs configure Services?
18) Dot net core 5.0 vs 6.0
19) Can we implement startup.cs file in 6.0?
20) What is dependency injection and types
21) What is service life cycle? (addTransient, addSingleton,addScoped), state thier difference
22) What is Routing and its types
23) What is an Attribute?
24) Serialization and Deserialization
25) How to implement Global exception?
26) What is session management?
27) What is caching and type of caching and how to implement it in dot net core?
28) Difference between Redis Cache and In-Memory Cache.
29) When do you use Distributed Caching?
30) Flow to update a single entity in Redis.
31) What is MVC pattern?

32) What is MVVP pattern?
33) What is CORS policy?
34) Why CORS no issue in postman but in browser.
35) What is Restful API?
36) Basic REST API principles.
37) SOAP vs REST
38) What is JSON?
39) JSON vs XML
40) POST vs PUT (Do not use records , idempotet)
41) POST vs PUT vs Patch.
42) If We can modify resource via Post call then why we need PUT call?
43) Can GET call have request body?
44) Can we add resource using GET call?
45) What is Content negotiations in Web API ?
46) Different Status Code and their usages.
47) Difference between 404 Not Found and 500 Internal Server Error.
48) If REST API is slow what will you do , where will you hunt , how to optimize ?
49) What is Api Versioning and its type? How to implement it
50) Return types in .net core?
51) Can we have multiple return types in net core?
52) Different return types in Web API.
53) Difference between Web API and Minimal API.
54) How to implement sorting and paging in .net core?
55) How to implement Authentication/Authorization in dot net core?
56) What is JWT? State its parts. How it is implemented?
57) How did you implement JWT logout.
58) When should we use Openid , Oauth as compared to Custom JWT ?
59) How to save sensitive data in database?
60) What are some common encryption techniques?
61) What data encryption techniques you have used in project?
62) Explain API Authorization (JWT / OAuth).
63) How do you revoke tokens with 30-minute expiry?
64) How do you invalidate tokens from UI?
65) If a user logs out from one browser, what happens in another browser session?
66) Encryption vs Hashing – differences and use cases.
67) How do you block specific IPs and where is it configured?
68) How would you design cloud-scale security for billions of users?
69) What is API gateway?
70) What is Rate limiter?
71) Suppose There is a requirement to upload a file with lakhs of records in database from mvc.
72) How to do it so transaction will be faster
73) Can you explain the MVC lifecycle and describe the flow of a request from the browser to the server and back?
74) What is cache management? How can it be implemented in a React front-end application? How do we manage caching in an ASP.NET MVC web application? What about in an ASP.NET Core Web API?

ORM

1) What is ORM?
2) Have you used any ORM in your project ? Which one?
3) What is Dapper?
4) What is EF core?
5) EF core vs framework?
6) What are navigation property in EF core?
7) How did you implement EF core with relational database?
8) Which approach did you follow in your project?
9) Code first vs Database first?
10) Commands used for code first
11) What are fluent APIs?
12) How do you call a Stored Procedure from EF core?
13) What is lazy loading
14) Eager loading in EF core
15) Tracking and Non Tracking in EF core
16) Can we handle transactions in EF core?
17) What is explicit transaction?

Azure

1. Different types of Azure services.
2. Examples of IaaS, PaaS, SaaS.
3. What is FaaS?
4. What are Azure Functions?
5. How do you deploy Azure Function Apps?
6. How do you deploy applications to cloud platforms?
7. What are the new features in recent .NET versions?
8. What is default namespace in .NET Core?

Devops CI CD Pipeline

1. Explain CI/CD pipeline.
2. How do you perform deployment without downtime?
3. Blue-Green vs Canary deployment.
4. How do you troubleshoot production issues?
5. Debugging strategies in live systems.

SQL Server and databases

1) What is Normalization and state its form?
2) What is BOYCE CODD normal form?
3) Indexes in SQL
4) Cluster vs Non cluster
5) What is execution plan?
6) Primary vs Unique
7) Union vs Union ALL
8) ISNULL vs Coalesce
9) Truncate vs Delete
10) Table vs Views
11) Stored Procedure vs Function
12) What is CTE?
13) Temporary table vs CTE
14) Temporary table vs Temporary Table variable
15) Sub Queries vs CTE
16) SQL Injection and how will you prevent it?
17) Triggers in sql, purpose?
18) What is Rownumber?
19) Rank vs DenseRank
20) What are Magic tables?
21) What is Deadlock?
22) What is WITH (NOLOCK)?
23) What is isolation level?
24) What is transaction and How to implement it?
25) Merged Statement
26) Send unlimited parameters through stored procedure
27) Difference between SQL vs NoSQL.
28) What is partitioning in NoSQL?
29) Explain primary key and secondary index.
30) How does SQL communication happen internally?
31) Find the Nth salary
32) You have a table named Students with columns StudentID, FirstName, LastName, and Score.Write an SQL query to find the names of students who have the same scores.

Architecture , Design Pattern and Microservices

1) Sometimes they tell to write code also , Singleton pattern code so try to attempt

2) What is Design Pattern and why do we need?
3) Types of Design Pattern - Creational, Structural, Behavioural
4) Which Design patterns you have used in in your project?
5) Which Design patterns you know?
6) What is Singleton DP? Type down the singleton pattern
7) Connection pooling many answer in singleton but that can be problematic. Especially from DB perspective.
8) Singleton pattern disadvantage
9) Breaking the Singleton using direct references. Iterator , IEnumerable
10) What is Strategy DP?
11) What is Adapter DP?
12) What is Abstract Factory DP?
13) What is Repository DP? list its advantages
14) What is CQRS DP?
15) What are 3 importanent elements
16) Why is mediatr in CQRS
17) Why CQRS is important in MS.
18) What is event sourcing in CQRS
19) Is SQL server Event source ?
20) Have you worked on Clean Architecture?
21) What are Microservices and its advantages and disadvantages?
22) Monolithic vs Microservices
23) Why do we use an API Gateway?
24) What is an API Gateway and how does it work?
25) Difference between rate limiting and load balancing.
26) Where should rate limiting be applied?
27) How do you prevent abuse using rate limiting?
28) How do microservices communicate?
29) How does RabbitMQ communication work?
30) Layer vs Tiered
31) Tier and N tier difference
32) How does one microservice call other microservice internally (Queues , WebAPI)
33) Strategic DDD vs Tactical DDD.
34) Whats a domain and sub domain ?
35) Types of sub domain
36) Bounded context , context map and UL.
37) What is a SAGA pattern?
38) What is DDD?
39) What are different classes , Entity , Value objects and Services
40) Aggregate root.
41) CQRS Pattern
42) Repository
43) Queues.
44) Draw infrastructure diagram
45) Project & Architecture (Authenticity Check)
46) Explain your current project end-to-end architecture.

47) Walk through the projects mentioned in your CV with technology justification.
48) What technologies were used and why?
49) Explain the system design of a large-scale platform (e.g., MakeMyTrip / Booking system).
50) Explain all major microservices in an online booking system.
51) Scenario-based questions on online booking system failures.
52) How do you avoid duplicate payments in a booking system?
53) How do you handle fault occurrence in an online booking system?
54) How can an API return the fastest search results in systems like MakeMyTrip?
55) List project layers in a large-scale application (UI, API, Services, Data, Infra).

AI ML

What is AI Training at a high level?

Difference between training vs inference.

SCENARIO BASED

1) How will you increase performance in angular application?
2) If there is case of bulk upload photos/videos, what will be your approch?
3) If you find slowness in angular application how will you optimize it?
4) **If an abstract Employee class has abstract method GetManager() and one concrete method GetSalary().**
   And Developer class and QA class inherits it.

   Now your Client said there is a requirement to add a CEO class which does not have a manager but has salary.

How you will implement this scenario?

Option - You can extend it as it is or you can change the existing code. But which one would be optimal?

5) There are 2 different Interfaces A, B and both are having same method name.
   Can a Class e.g. class C, implement both of them?
   If yes, then how will it use the method from Class A.

6) There is a sealed class and I want to add new methods to it without changing the existing logic.

7) Can a class have multiple constructor with different access modifiers also with static constructor, if yes then in what order it will execute?

8) If you need to fetch data but parameter/query string is complex then you will use GET or POST call and why?
9) How to increase performance in dot net core?
10) How to do bulk insert, update in dot net core?
11) How to limit the API hits in dot net core?
12) How to handle huge payloads that expires in 30 secs?
13) In case of complex logic, will you use linq (dot net core) or SQL (database)?
14) How to optimize a Stored procedure?
15) If we told you to create a new project then by which approach you will choose Code first or Database first and why?
16) How to do bulk insert, update in EF core?

Unit / API Testing

1) Why we need testing?
2) What is Unit testing?
3) What is Integration testing?
4) Difference Unit and Integration?
5) Which unit testing framework you used in your project?
6) What is the code coverage percentage?
7) What is Moq. Have you used it in project?
8) How to handle exception in unit testing?

CODING QUESTIONS

Many companies tell code in fiddlenet

1) Finding two numbers in an array that add up to a target
2) Reverse word in a string
3) Reverse Array
4) Find occurrence of each character in a given string
5) Move 0's to end of the array
6) Find the First Non-Repeating Character
7) Remove Duplicates from a String
8) Reverse a String
9) Swapping two number

10) Extract the surname
11) DSA based question
12) Check if a String is a Palindrome
13) Find the Longest Substring Without Repeating Characters
14) Remove All White Spaces from a String
15) Find the Longest Palindromic Substring
16) Can you write a code on Boxing and Unboxing in C#?
17) . int[] GivenArray = { 1, 2, 3, 4, 5, 6 };
    //Print Output-> //1 2 3 4//5 6 1 2
18) Left/Right rotate an array by K places.
19) Prime numbers
20) Factorial using recursion
21) **How can we return multiple data with different data types from a function in .c#?**
22) Find max number in Array.
23) Take the array of numbers stored in arr and return the string true if any combination of numbers in the array (excluding the largest number) can be added up to equal the largest number in the array, otherwise return the string false. For example: if arr contains [4, 6, 24, 10, 1, 3] the output should return true because 4 + 6 + 10 + 3 + 1 = 24. The array will not be empty, will not contain all the same elements, and may contain negative numbers.

Examples

Input: new int[] {5,7,16,1,2}
Output: false
Input: new int[] {3,5,-1,8,15}
Output: true

24) Fibonical series - output = 1, 1, 2, 3, 5, 8, 13, 21
25) Basic Star pattern examples
26) Question related to string - Reverse, SubString, IndexOf, Contains, Replace, Split etc
27) How many times letter has occured ?

Input - hello world

Output -

h – 1
e – 1
l – 3
o – 2
w – 1
r – 1
d - 1

28) input - ab+cde+**
    output - ((a+b)(c(d+e)))

29) They will give a snippet and we need to Spot errors and correct it.
30) They will give a snippet and we need to optimize it or refactor it.

31) Questions related to DSA e.g. Stack and Queue
32) What is the next number: 1, 3, 7, 13----
33) Create a countdown timer that will countdown from 30 and update every second.
When the countdown hits 0, display 'Take Off',
There is a Reset button next to the timer.
The Reset button will reset the timer to 30 sec if the takeoff has not already happened.
After takeoff, reset is not possible.

34) Write example of Date Pipe with short date format
35) Find Duplicate records in table
36) Find Nth highest Salary
37) Find count of employees having 2nd highest salary in each department.
38) Find the Employee's Manager in the same table.
39) select Count(*) vs select Count(name) vs select Count(address). Which one is faster?
40) Find max salary from employees
41) Find employees having 2nd highest salary
42) Left join in LINQ with any Method syntax or Query syntax
1) Anagram code
2) Remove duplicate from list
3) Reverse string

1.Reverse a String

2. Check if a String is a Palindrome

3. Remove Duplicates from a String

4. Find the First Non-Repeating Character

5. Count the Occurrences of Each Character

6. Reverse Words in a Sentence

7. Check if Two Strings are Anagrams

8. Find the Longest Substring Without Repeating Characters

9. Convert a String to an Integer (atoi Implementation)

10. Compress a String (Run-Length Encoding)

11. Find the Most Frequent Character

12. Find All Substrings of a Given String

13. Check if a String is a Rotation of Another String

14. Remove All White Spaces from a String

15. Check if a String is a Valid Shuffle of Two Strings

16. Convert a String to Title Case

17. Find the Longest Common Prefix

18. Convert a String to a Character Array

19. Replace Spaces with %20 (URL Encoding)

20. Convert a Sentence into an Acronym

21. Check if a String Contains Only Digits

22. Find the Number of Words in a String

23. Remove a Given Character from a String

24. Find the Shortest Word in a String

25. Find the Longest Palindromic Substring

4) code –

```
const object = {   who: 'World',   greet() {     return Hello, ${this.who}!;   }, farewell: () => {     return Goodbye, ${this.who}!;} }; console.log(object.greet());    // What is logged?
console.log(object.farewell()); // What is logged?
```

=> Format the above code and write its output.

5) Code -

```
(function immediateA(a) {   return (function immediateB(b) {     console.log(a);   })(1); })(0); //What is logged on console?
```

=> Format the above code and write its output.

6) code –

```
const fruits=["jackfruit","mango","apple"];
```

=> Can we modify the above array, if yes then replace apple with banana?

7) code -

```
a= 10
var a;
console.log(a)
```

=> output?

8) Write a function to convert a string from camel case to snake case.

for e.g input : thisIsATest

output : This_Is_A_Test

9) You are given a nested array, you have to write a function to flatten it.
For eg. 1. Input - [1, [2]] Output - [1,2]

10) Give output for below variables
let a = '3' + 3 + 3 = ?
let b = 3 + '3' + 3 = ?
let c = 3 + 3 + '3' = ?
let d = '3' + 3 - 3 = ?
let e = 3 + 3 + 3 = ?